

Engineering Mechanics

Lecture – III

3 Dimensional Force Systems

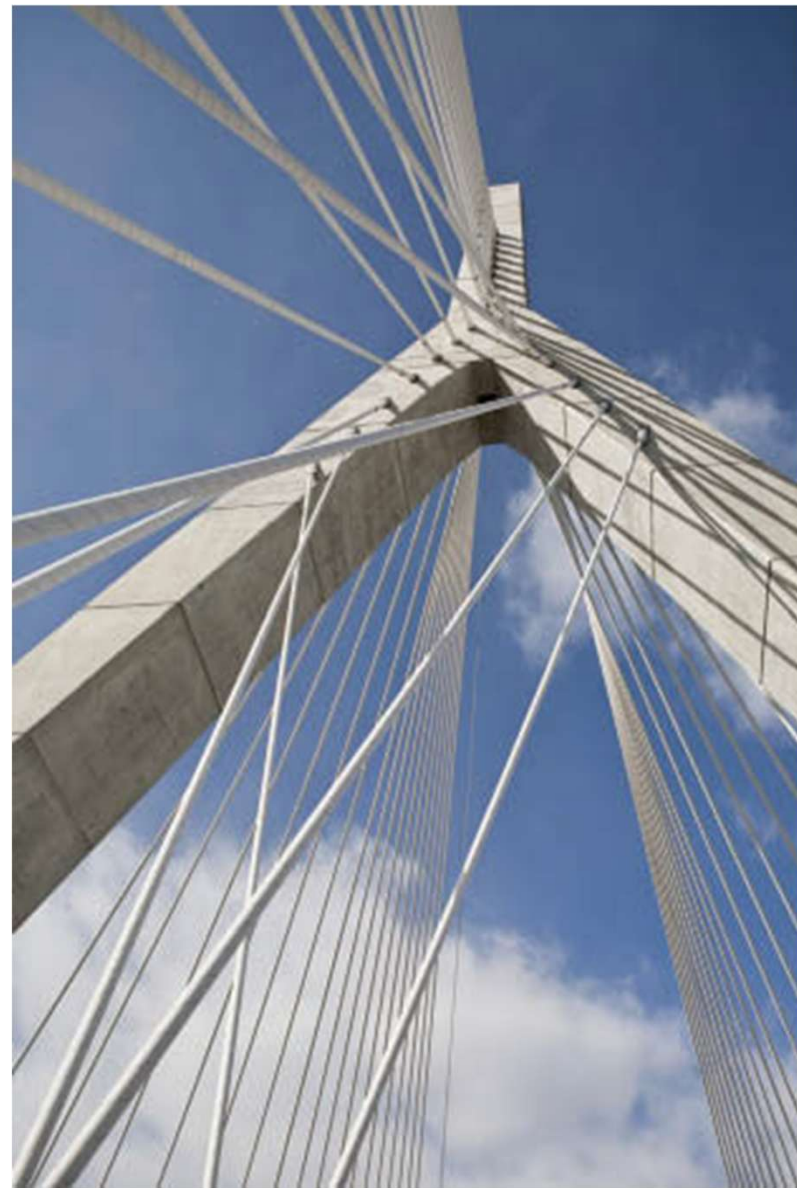
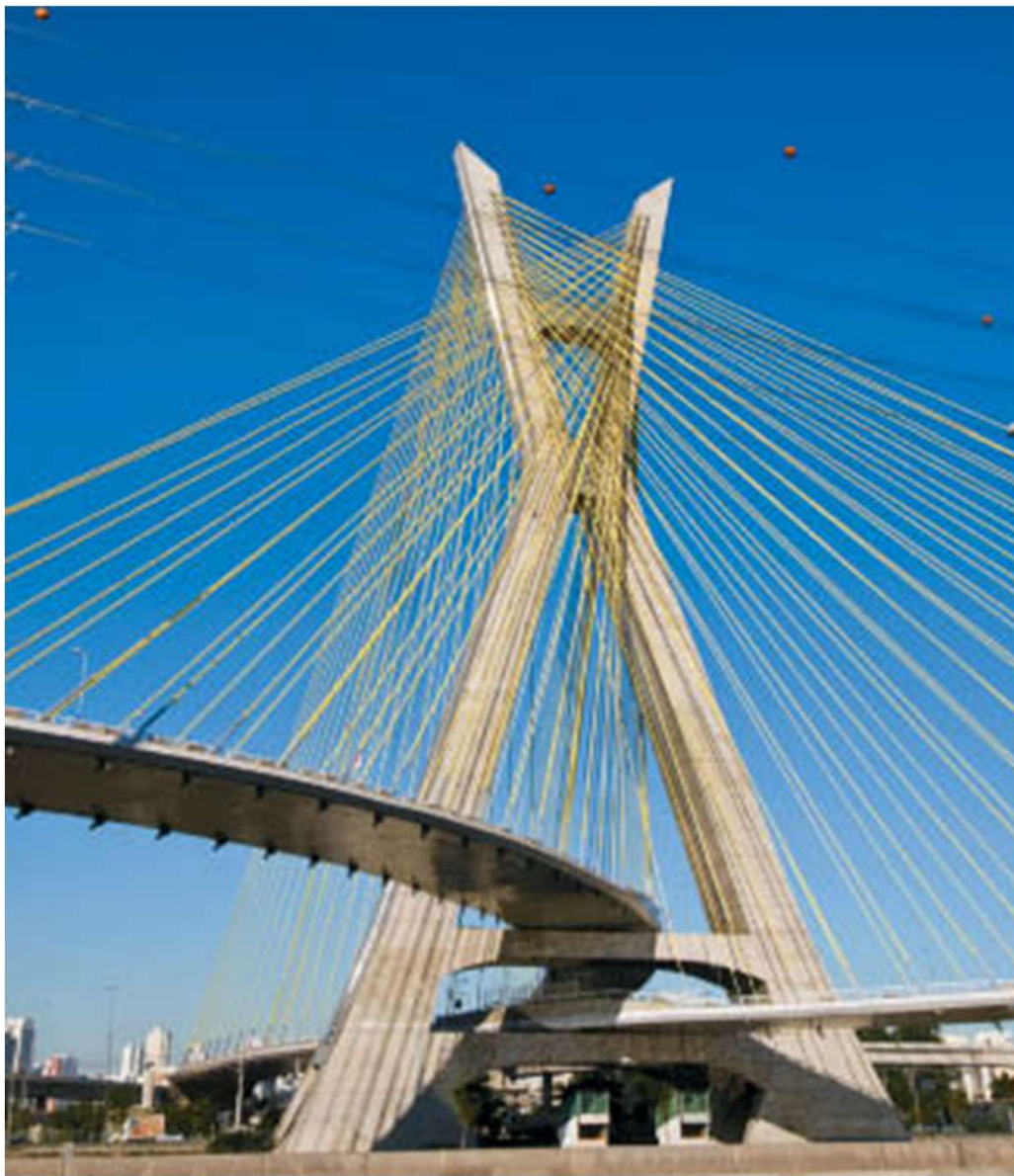
By

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RECTANGULAR COMPONENTS

Direction cosine representation

$$F_x = F \cos \theta_x$$

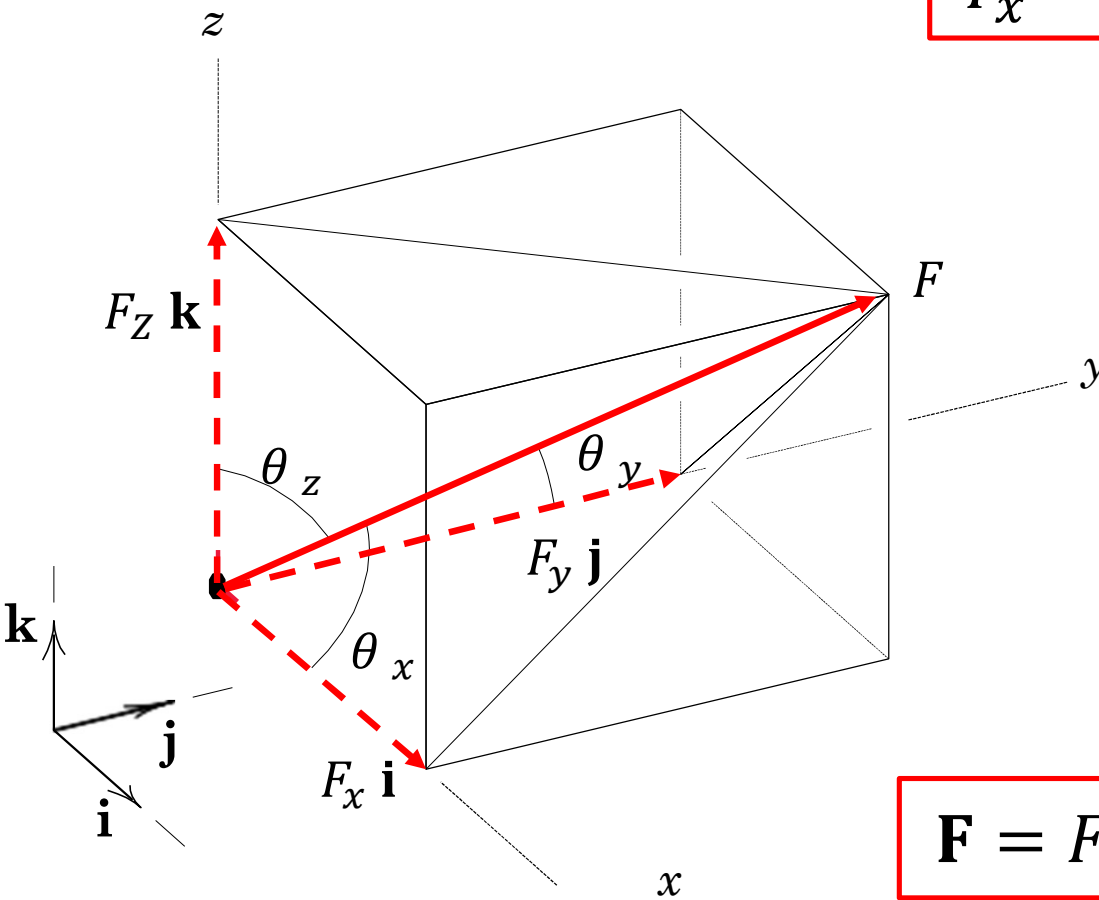
$$F_y = F \cos \theta_y$$

$$F_z = F \cos \theta_z$$

$$F = \sqrt{F_x^2 + F_y^2 + F_z^2}$$

$$\mathbf{F} = F_x \mathbf{i} + F_y \mathbf{j} + F_z \mathbf{k}$$

$$\mathbf{F} = F(\cos \theta_x \mathbf{i} + \cos \theta_y \mathbf{j} + \cos \theta_z \mathbf{k})$$



RECTANGULAR COMPONENTS

$$\mathbf{F} = F(\cos \theta_x \mathbf{i} + \cos \theta_y \mathbf{j} + \cos \theta_z \mathbf{k})$$

$$\mathbf{F} = F(l\mathbf{i} + m\mathbf{j} + n\mathbf{k})$$

$$l = \cos \theta_x$$

$$m = \cos \theta_y$$

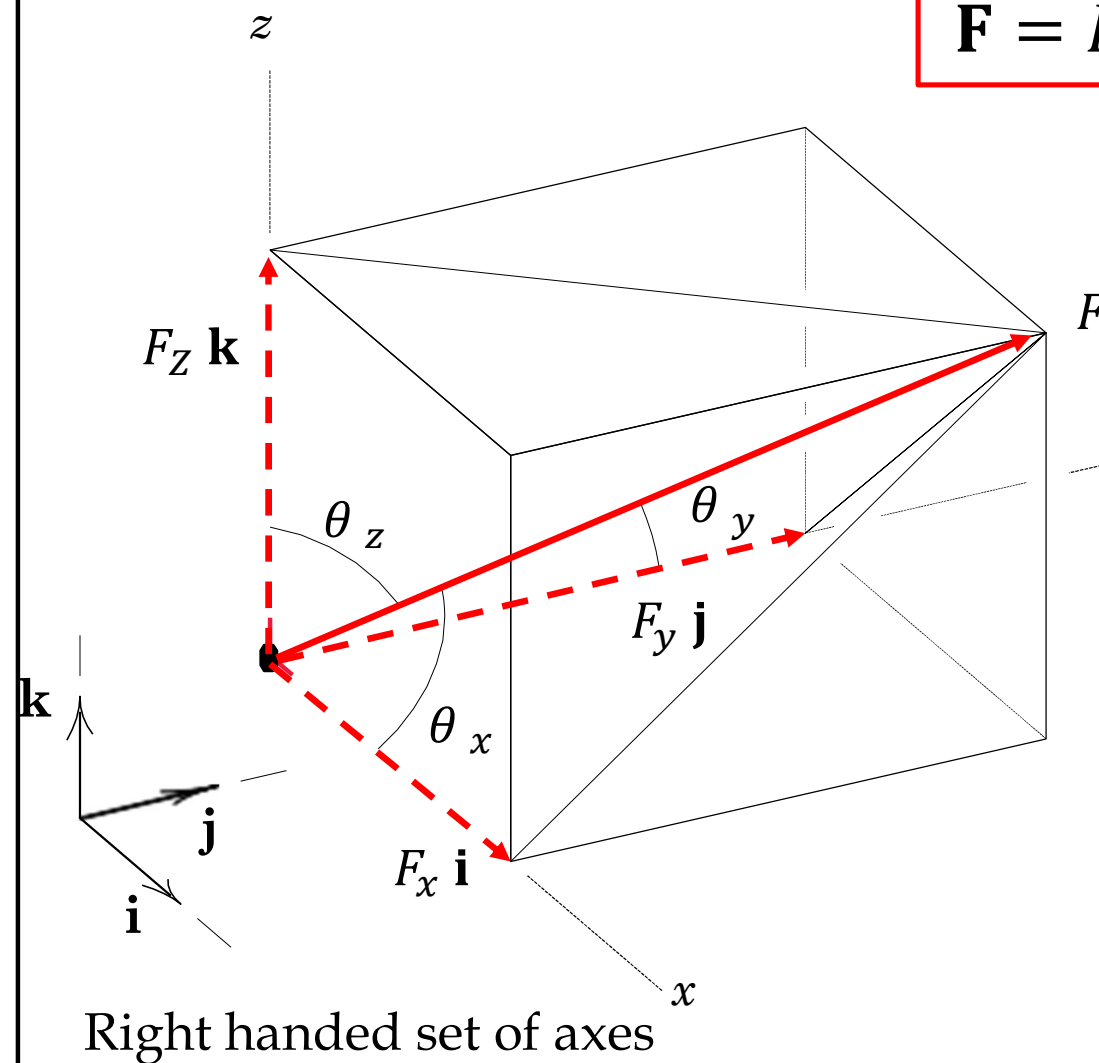
$$n = \cos \theta_z$$

$$l^2 + m^2 + n^2 = 1$$

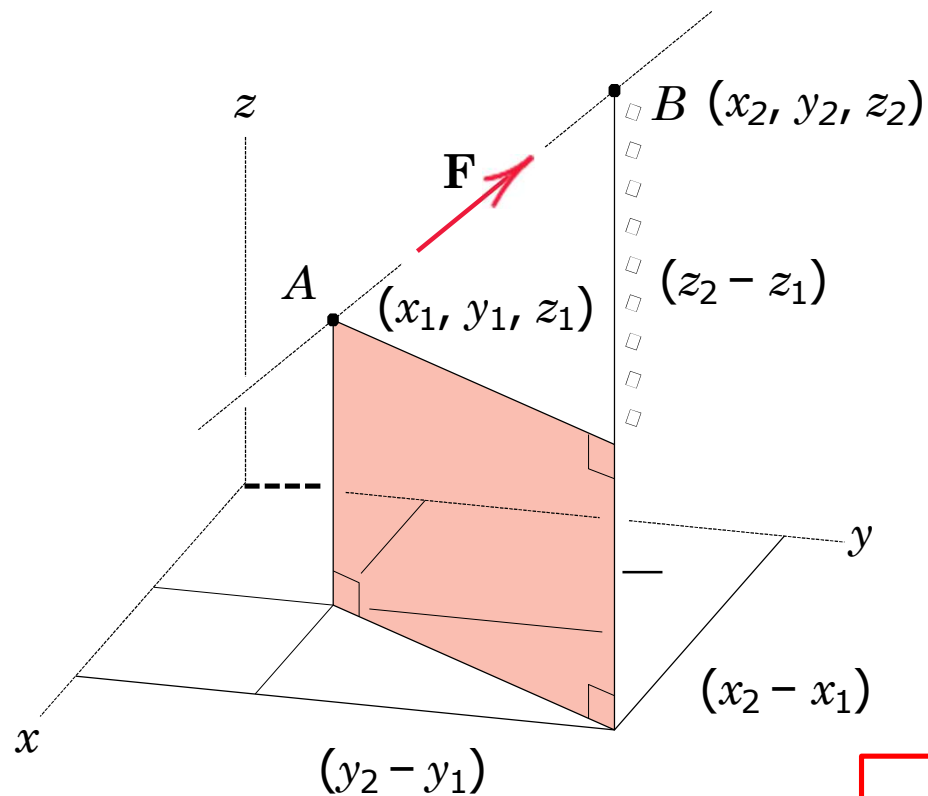
Force in terms of unit vector

$$\mathbf{F} = F\mathbf{n}_F$$

$$\mathbf{n}_F = (l\mathbf{i} + m\mathbf{j} + n\mathbf{k})$$



Direction Specification by two points

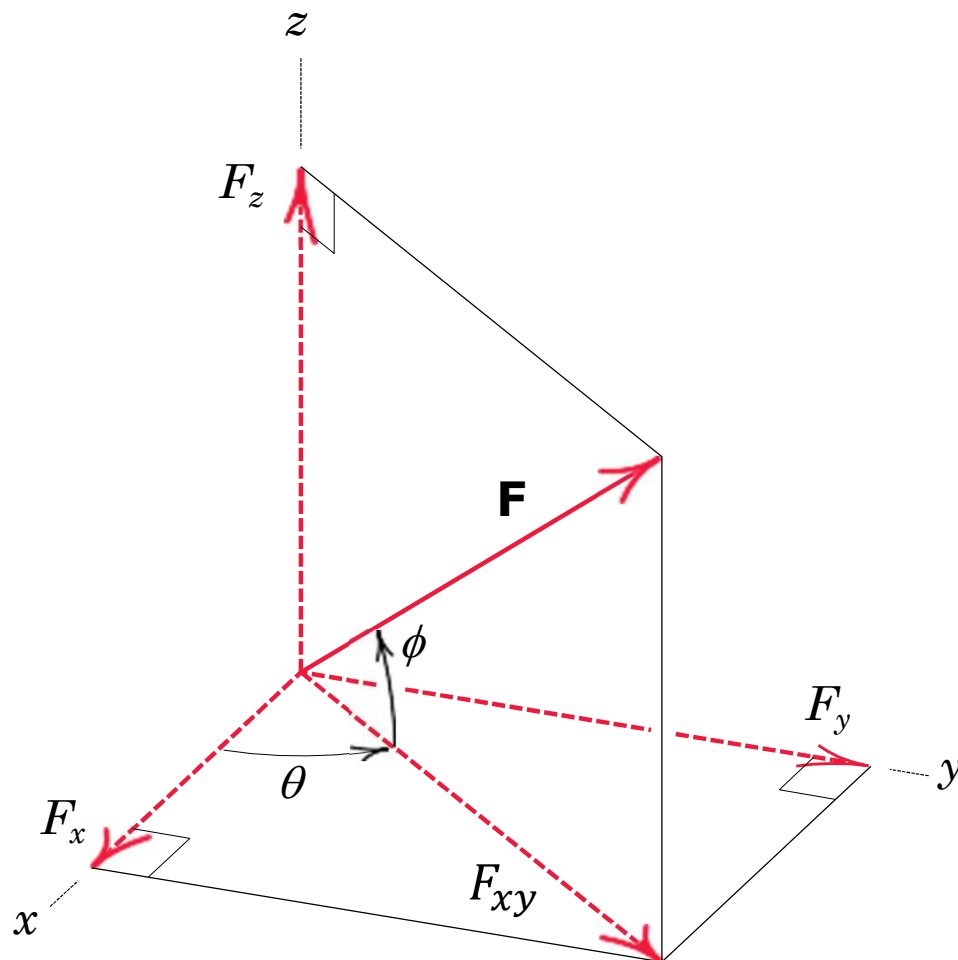


$$\mathbf{F} = F \mathbf{n}_F$$

$$\mathbf{n}_F = \frac{\overrightarrow{AB}}{\overline{AB}}$$

$$\mathbf{n}_F = \frac{(x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} + (z_2 - z_1)\mathbf{k}}{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}}$$

Direction Specification by two angles



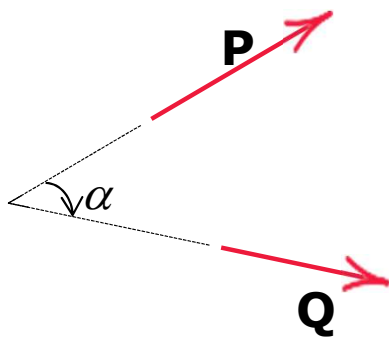
$$\mathbf{F} = F_x \mathbf{i} + F_y \mathbf{j} + F_z \mathbf{k}$$

$$F_z = F \sin \varphi$$

$$F_{xy} = F \cos \varphi$$

$$F_x = F_{xy} \cos \theta = F \cos \varphi \cos \theta$$

$$F_y = F_{xy} \sin \theta = F \cos \varphi \sin \theta$$

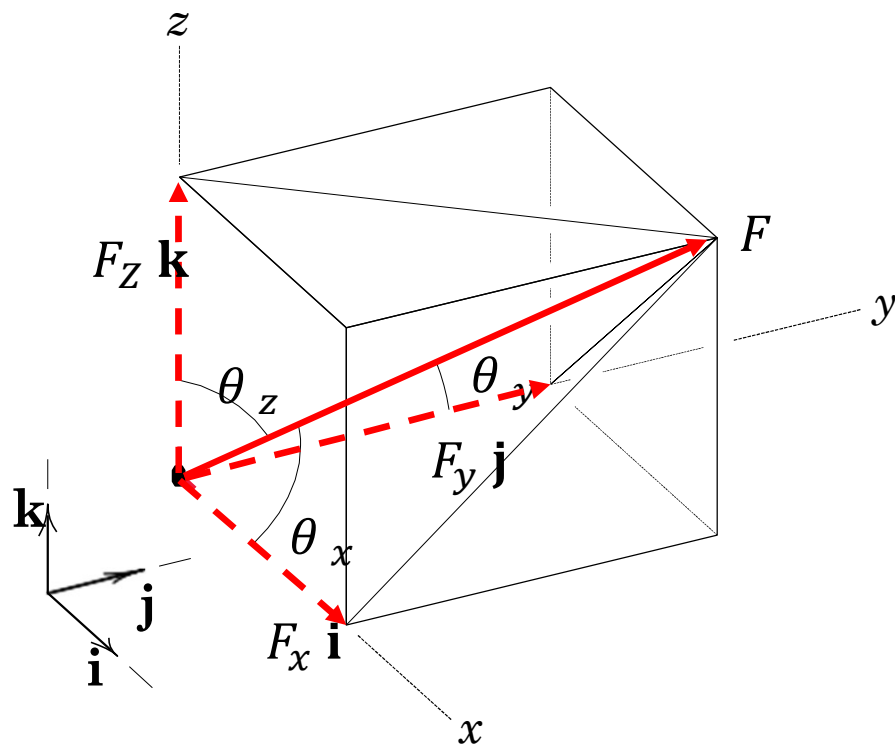


Dot Product

$$\mathbf{P} \cdot \mathbf{Q} = (P \cos \alpha) Q$$

$$\mathbf{P} \cdot \mathbf{Q} = PQ \cos \alpha$$

Projection of $\mathbf{F}$ along x



$$\mathbf{F} \cdot \mathbf{i} = F \cos \theta_x$$

Projection of Force (F)

Projection of F along the axis defined by unit vector $\mathbf{n}$

$$\mathbf{F} = F(l\mathbf{i} + m\mathbf{j} + n\mathbf{k})$$

$$\mathbf{n} = \alpha\mathbf{i} + \beta\mathbf{j} + \gamma\mathbf{k}$$

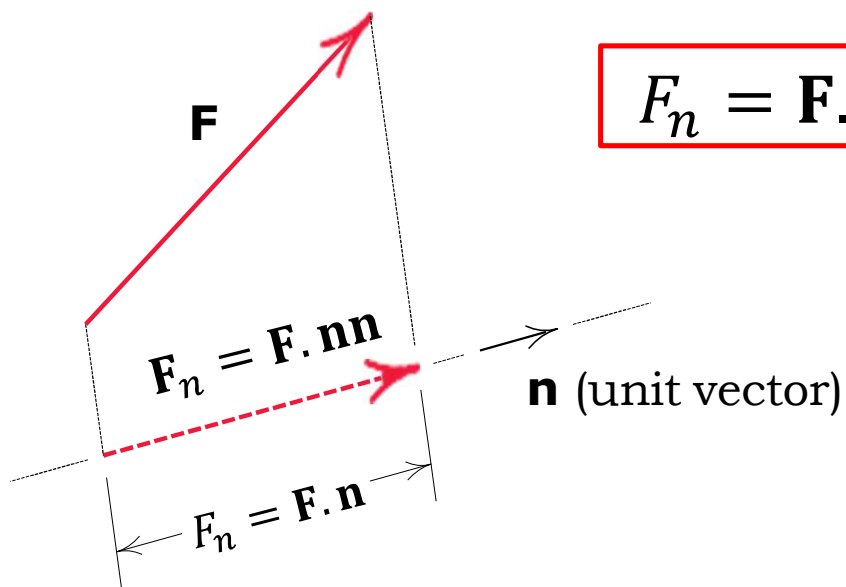
$$F_n = \mathbf{F} \cdot \mathbf{n} = F(l\mathbf{i} + m\mathbf{j} + n\mathbf{k}) \cdot (\alpha\mathbf{i} + \beta\mathbf{j} + \gamma\mathbf{k})$$

$$F_n = F(l\alpha + m\beta + n\gamma)$$

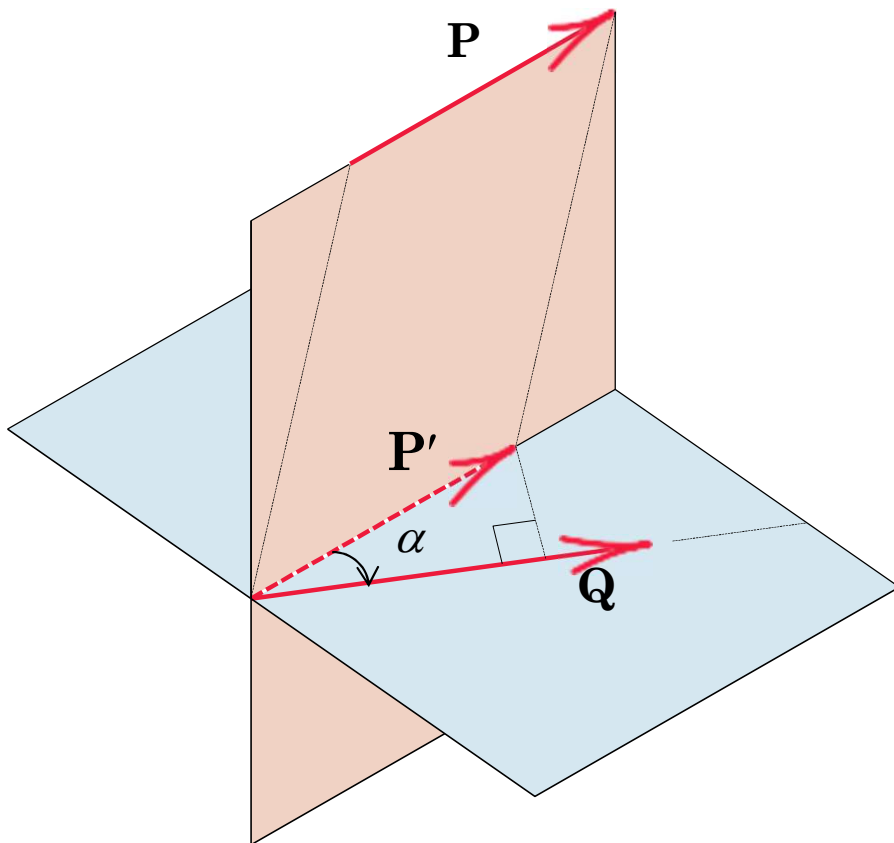
Since

$$\mathbf{i} \cdot \mathbf{i} = \mathbf{j} \cdot \mathbf{j} = \mathbf{k} \cdot \mathbf{k} = 1$$

$$\mathbf{i} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{i} = \mathbf{i} \cdot \mathbf{k} = \mathbf{k} \cdot \mathbf{i} = \mathbf{k} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{k} = 0$$



Angle between Two Vectors



$$\mathbf{P} \cdot \mathbf{Q} = PQ \cos \alpha$$

$$\alpha = \cos^{-1} \frac{\mathbf{P} \cdot \mathbf{Q}}{PQ}$$

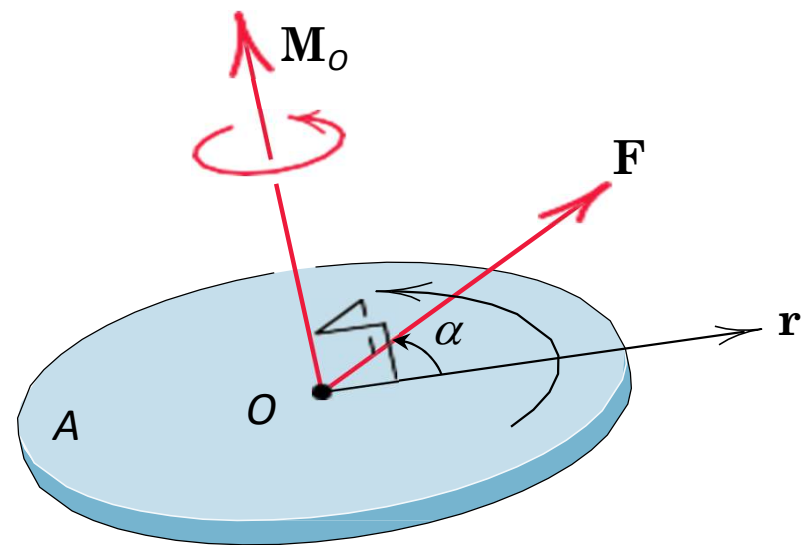
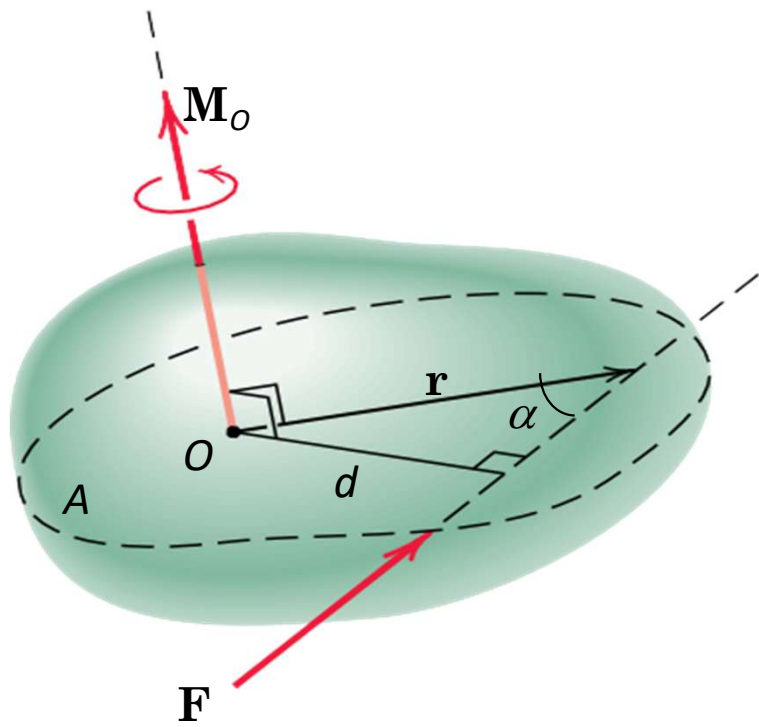
This also true for P'

$$\theta = \cos^{-1} \frac{\mathbf{F} \cdot \mathbf{n}}{F}$$

Moment in 3 Dimensions

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$$

$\mathbf{r}$: vector from O to any point on line of action of $\mathbf{F}$



Evaluating the Cross Product

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$$

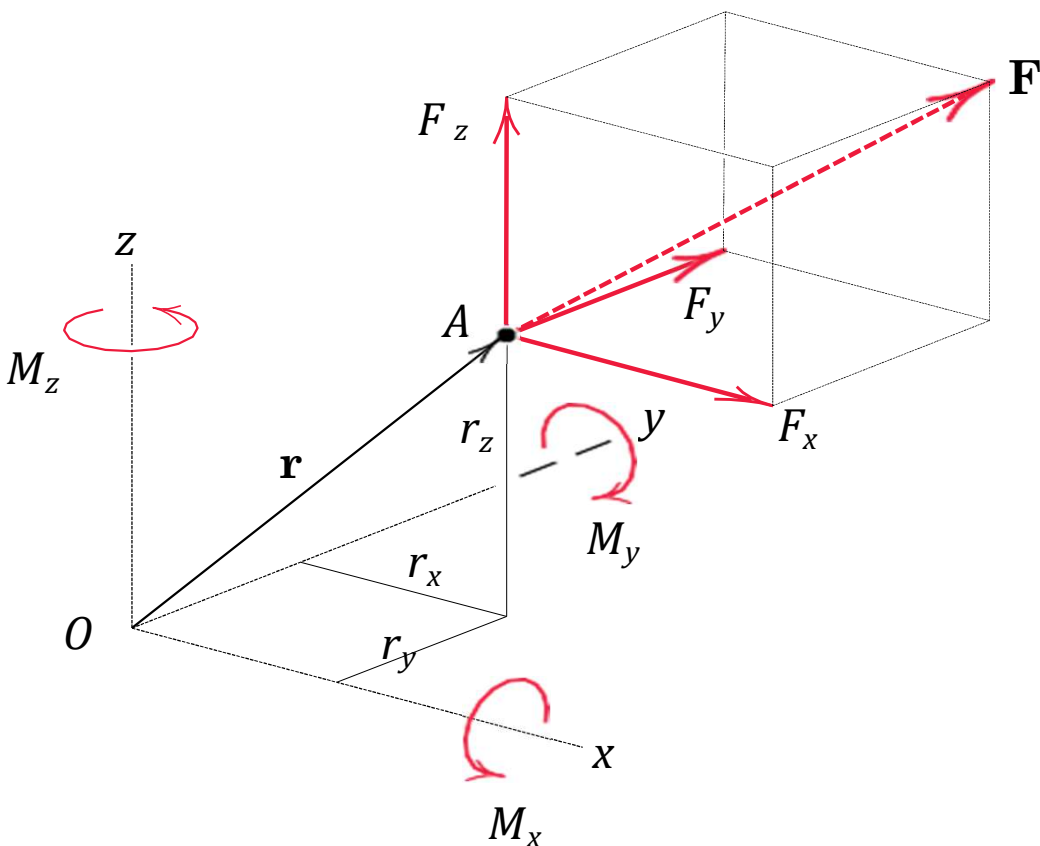
$$\mathbf{M}_O = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix}$$

$$\mathbf{M}_O = (r_y F_z - r_z F_y) \mathbf{i} + (r_z F_x - r_x F_z) \mathbf{j} + (r_x F_y - r_y F_x) \mathbf{k}$$

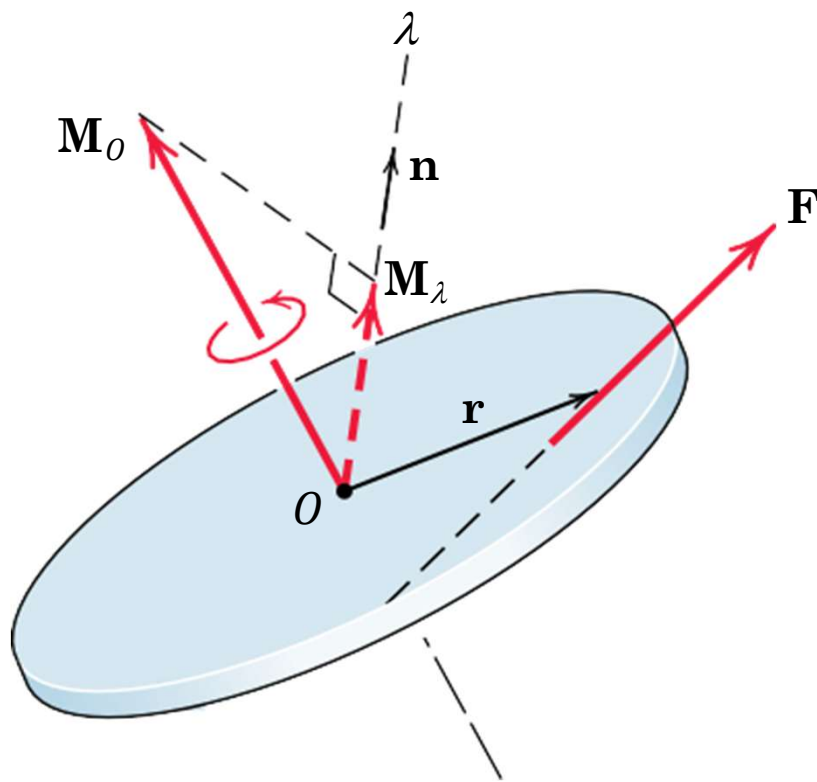
$$M_x = (r_y F_z - r_z F_y)$$

$$M_y = (r_z F_x - r_x F_z)$$

$$M_z = (r_x F_y - r_y F_x)$$



Moment about an Arbitrary Axis

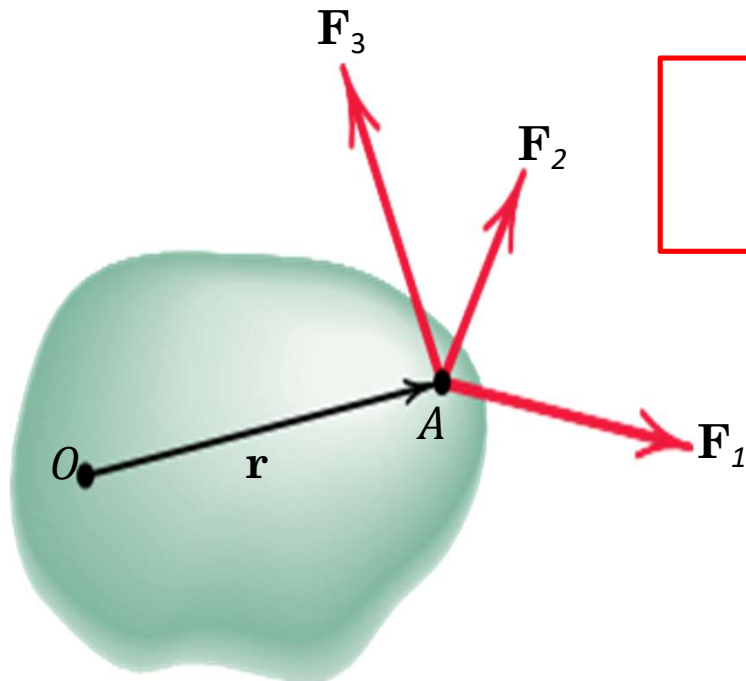


$$\mathbf{M}_\lambda = [(\mathbf{r} \times \mathbf{F}) \cdot \mathbf{n}] \mathbf{n}$$

Triple Scalar product

$$|\mathbf{M}_\lambda| = M_\lambda = \begin{vmatrix} r_x & r_y & r_z \\ F_x & F_y & F_z \\ \alpha & \beta & \gamma \end{vmatrix}$$

Varignon's Theorem in Three Dimensions



$$\mathbf{r} \times \mathbf{F}_1 + \mathbf{r} \times \mathbf{F}_2 + \mathbf{r} \times \mathbf{F}_3 + \dots \\ = \mathbf{r} \times (\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \dots) = \mathbf{r} \times \Sigma \mathbf{F}$$

$$\mathbf{M}_O = \Sigma(\mathbf{r} \times \mathbf{F}) = \mathbf{r} \times \mathbf{R}$$

Couples in Three Dimensions

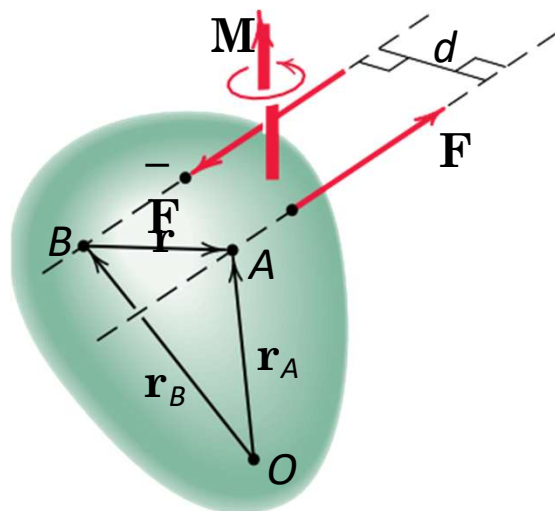
Taking moment of $\mathbf{F}$ and $-\mathbf{F}$ about point O

$$\mathbf{M}_O = \mathbf{r}_A \times \mathbf{F} + \mathbf{r}_B \times (-\mathbf{F})$$

$$\mathbf{M}_O = (\mathbf{r}_A - \mathbf{r}_B) \times \mathbf{F}$$

$$\mathbf{r} = (\mathbf{r}_A - \mathbf{r}_B)$$

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$$

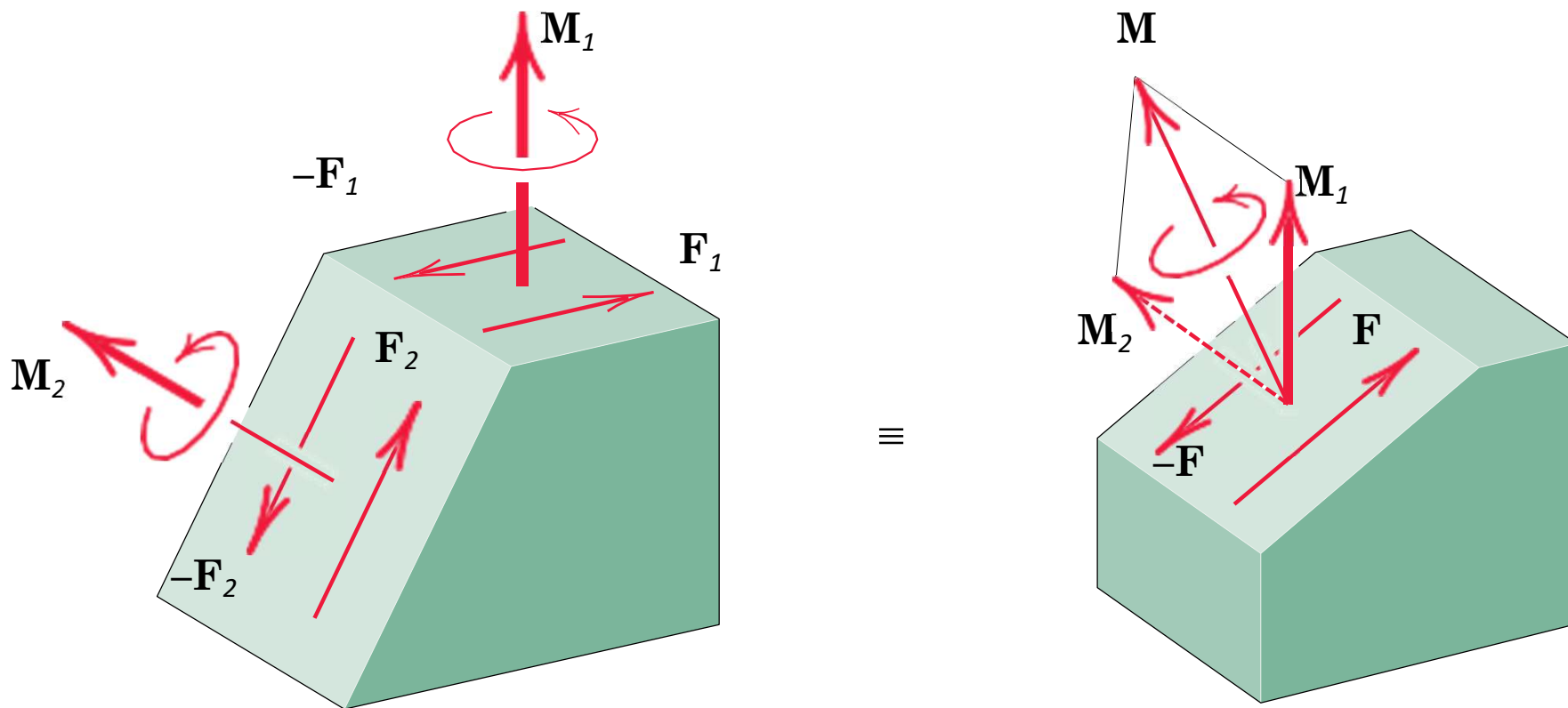


$\mathbf{r}$: vector from any point on line of action of $-\mathbf{F}$ to any point on line of action of $\mathbf{F}$

Independent from O and hence free vector

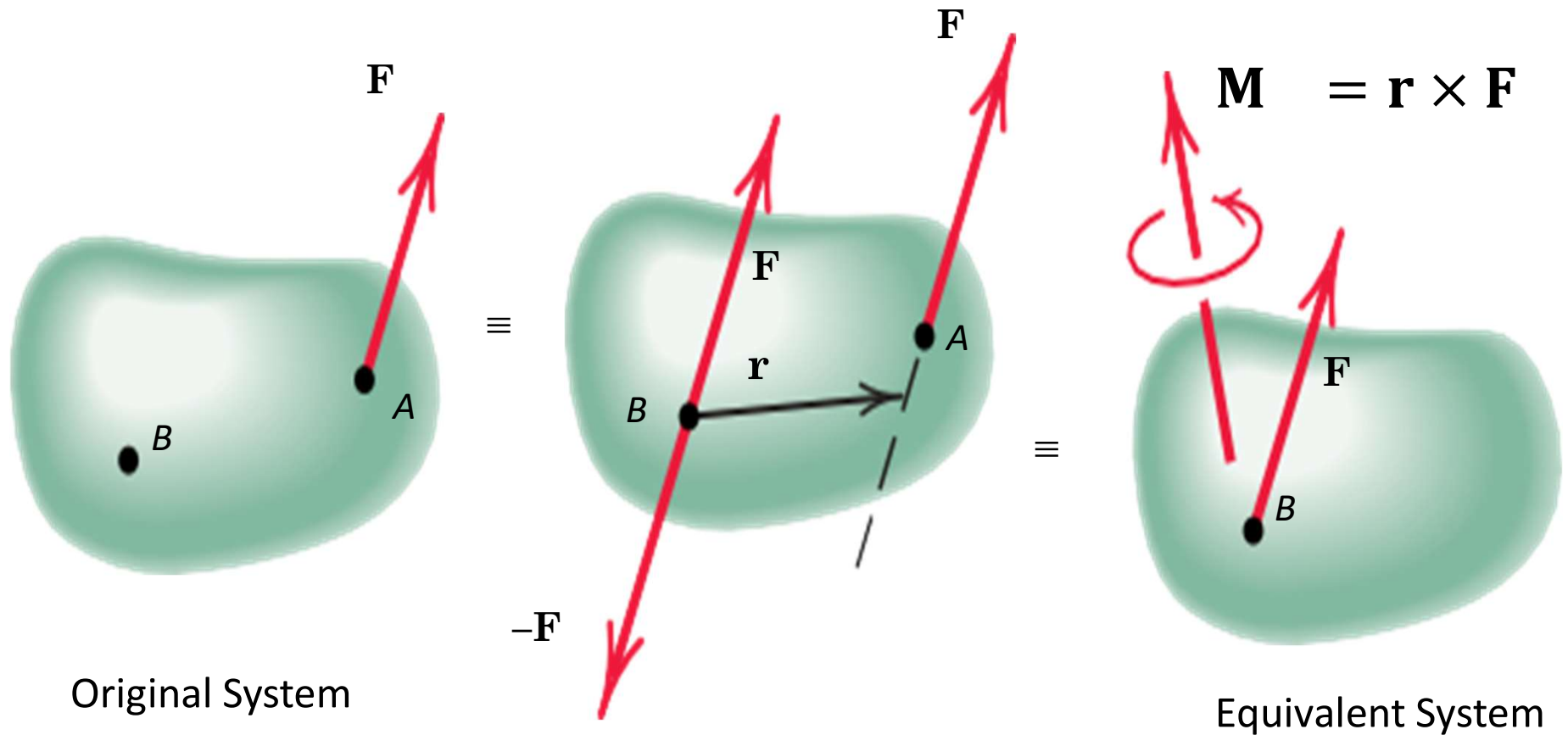
Note: The moment of a couple is a *free vector*, whereas the moment of a force about a point is a *sliding vector* whose direction is along the axis through the point.

Summation of Couples

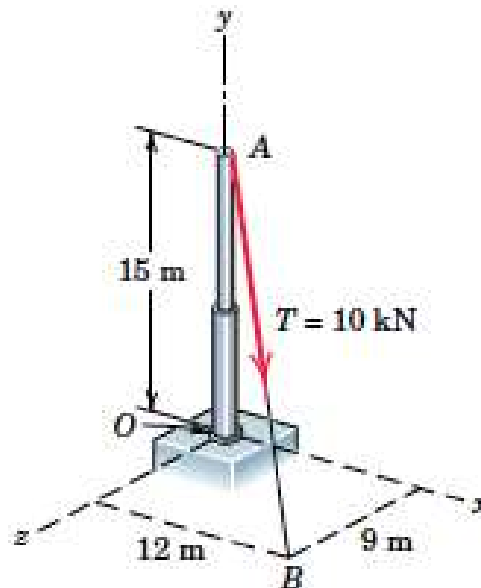


All rule of vector addition apply

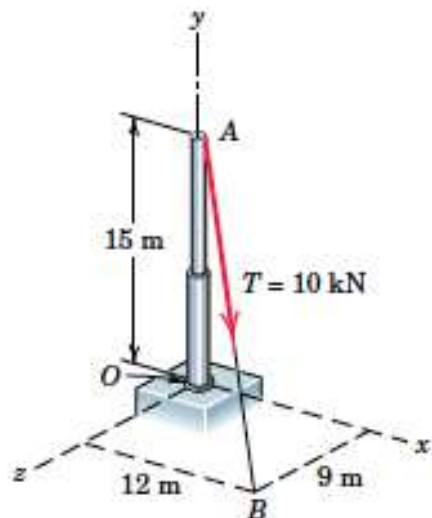
Force-couple system in D



- Example 1: A tension T of magnitude 10 kN is applied to the cable attached to the top A of the rigid mast and secured to the ground at B . Determine the moment M_z of T about the z -axis passing through the base O .



Solution:



$$\mathbf{T} = T\mathbf{n}_{AB} = 10 \left[\frac{12\mathbf{i} - 15\mathbf{j} + 9\mathbf{k}}{\sqrt{(12)^2 + (-15)^2 + (9)^2}} \right]$$

$$= 10(0.566\mathbf{i} - 0.707\mathbf{j} + 0.424\mathbf{k}) \text{ kN}$$

$$[\mathbf{M}_O = \mathbf{r} \times \mathbf{F}]$$

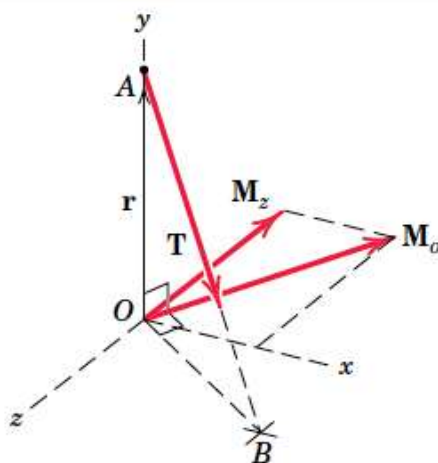
$$\mathbf{M}_O = 15\mathbf{j} \times 10(0.566\mathbf{i} - 0.707\mathbf{j} + 0.424\mathbf{k})$$

$$= 150(-0.566\mathbf{k} + 0.424\mathbf{i}) \text{ kN} \cdot \text{m}$$

$$M_z = \mathbf{M}_O \cdot \mathbf{k}$$

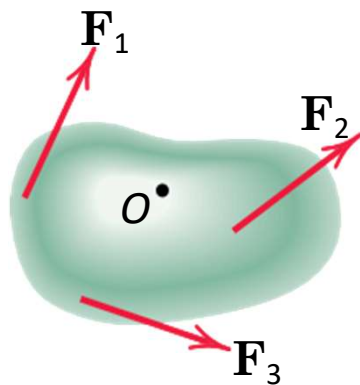
$$M_z = 150(-0.566\mathbf{k} + 0.424\mathbf{i}) \cdot \mathbf{k}$$

$$M_z = -84.9 \text{ kN} \cdot \text{m}.$$

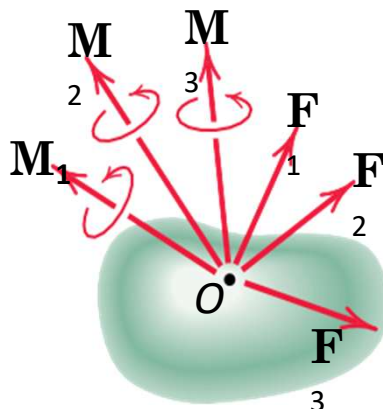


Resultant

Original System

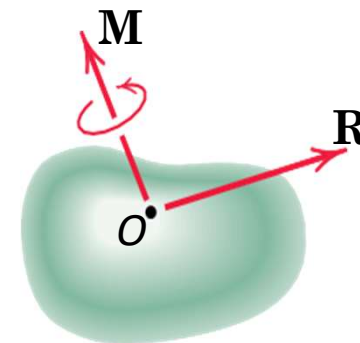


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Equivalent System



$$\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \cdots = \Sigma \mathbf{F}$$

$$\mathbf{M} = \mathbf{M}_1 + \mathbf{M}_2 + \mathbf{M}_3 + \cdots = \Sigma(\mathbf{r} \times \mathbf{F})$$

$$R_x = \Sigma F_x$$

$$R_y = \Sigma F_y$$

$$M_x = \Sigma(\mathbf{r} \times \mathbf{F})_x$$

$$M_y = \Sigma(\mathbf{r} \times \mathbf{F})_y$$

$$R_z = \Sigma F_z$$

$$M_z = \Sigma(\mathbf{r} \times \mathbf{F})_z$$

$$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2 + (\Sigma F_z)^2}$$

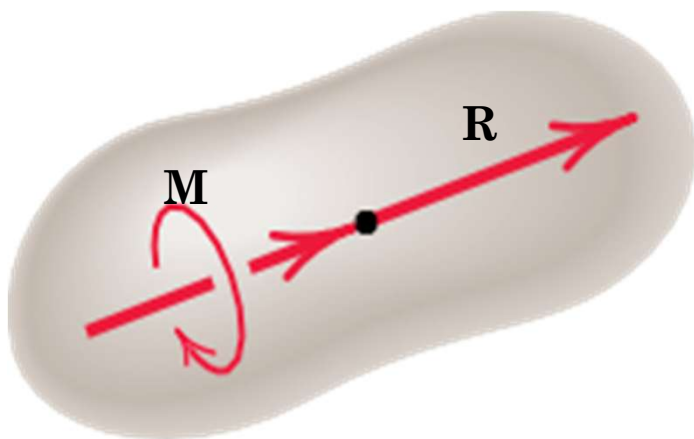
$$M = \sqrt{M_x^2 + M_y^2 + M_z^2}$$

Special Cases

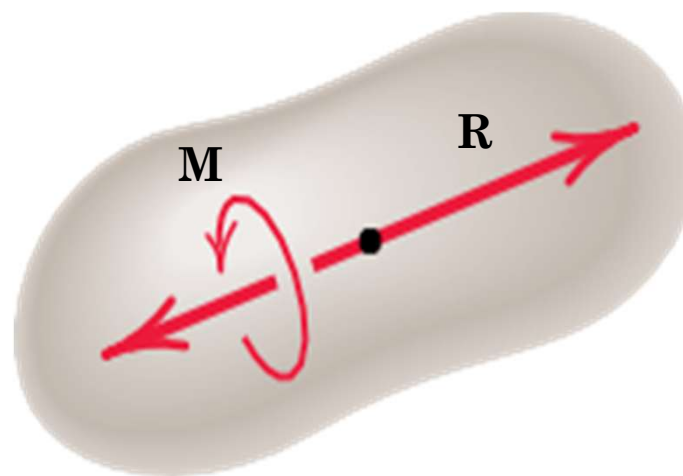
- **Concurrent Forces**
 - No moments about the point of concurrency.
- **Parallel Forces**
 - Magnitude of the parallel resultant force $\mathbf{R}$ is simply the magnitude of the algebraic sum of the given forces.
- **Coplanar Forces**
 - Similar to 2D cases

Wrench Resultant

- When the resultant couple vector M is parallel to the resultant force R , the resultant is called a wrench

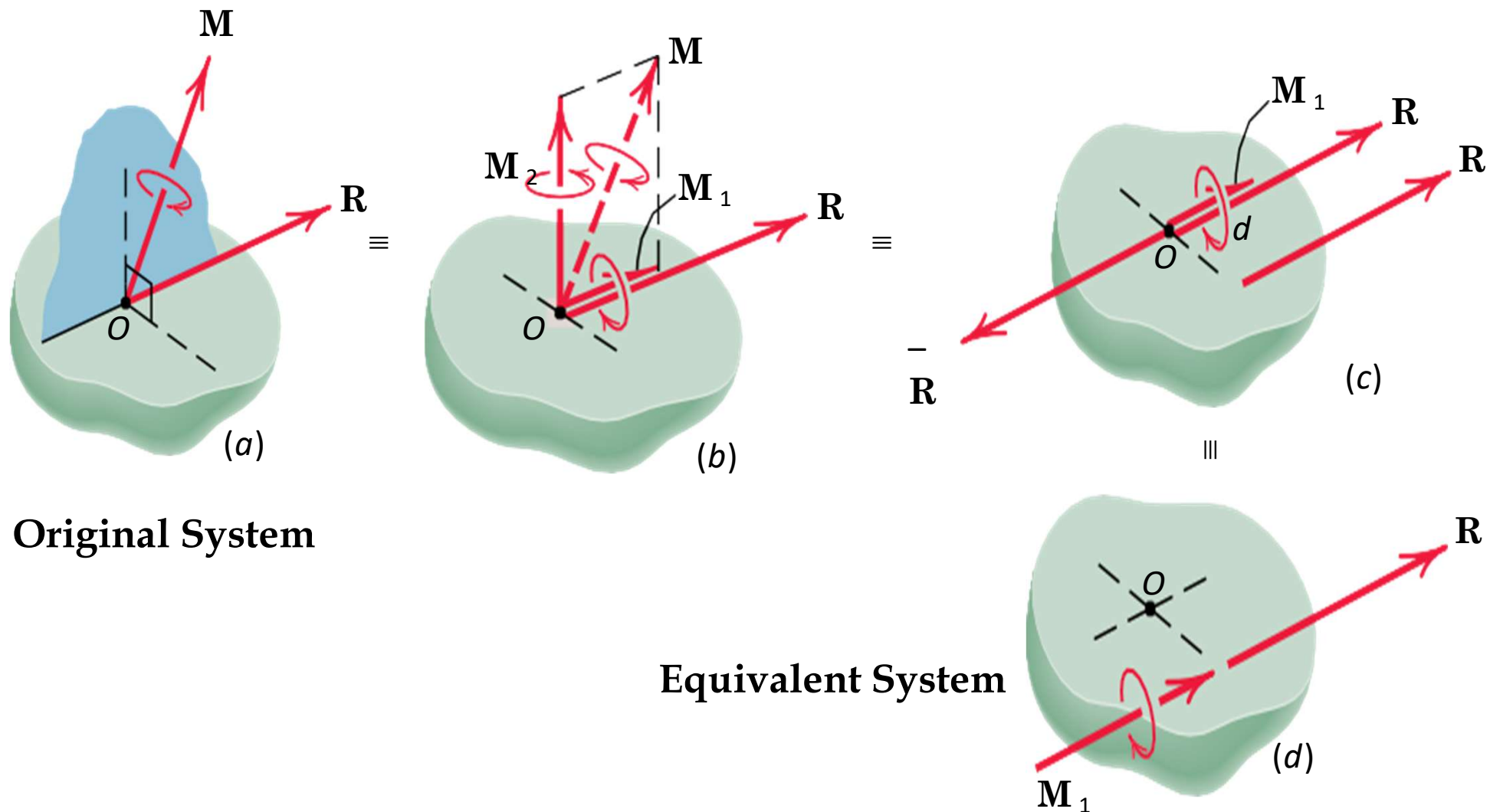


Positive wrench

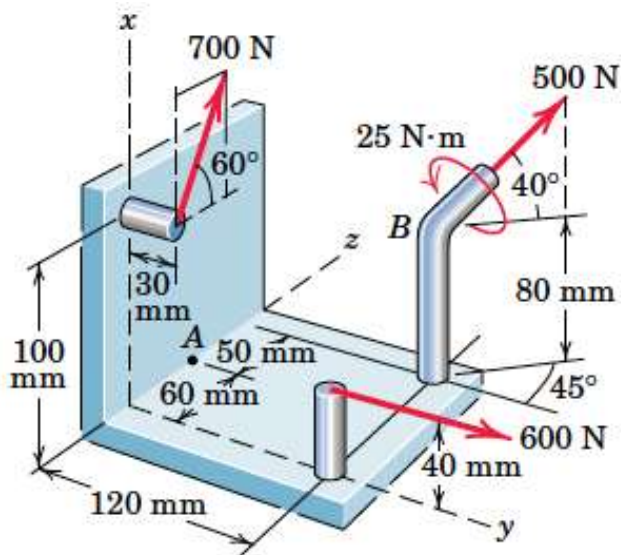


Negative wrench

Wrench Resultant



- Example 2: Replace the two forces and the negative wrench by a single force R applied at A and the corresponding couple M .



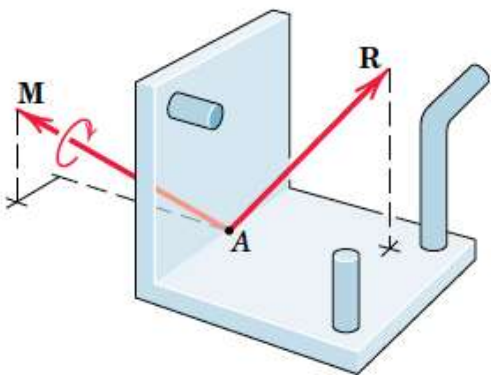
$$[R_x = \Sigma F_x] \quad R_x = 500 \sin 40^\circ + 700 \sin 60^\circ = 928 \text{ N}$$

$$[R_y = \Sigma F_y] \quad R_y = 600 + 500 \cos 40^\circ \cos 45^\circ = 871 \text{ N}$$

$$[R_z = \Sigma F_z] \quad R_z = 700 \cos 60^\circ + 500 \cos 40^\circ \sin 45^\circ = 621 \text{ N}$$

$$\mathbf{R} = 928\mathbf{i} + 871\mathbf{j} + 621\mathbf{k} \text{ N}$$

$$R = \sqrt{(928)^2 + (871)^2 + (621)^2} = 1416 \text{ N}$$



Example 2

$$[\mathbf{M} = \mathbf{r} \times \mathbf{F}]$$

$$\mathbf{M}_{500} = (0.08\mathbf{i} + 0.12\mathbf{j} + 0.05\mathbf{k}) \times 500(\mathbf{i} \sin 40^\circ + \mathbf{j} \cos 40^\circ \cos 45^\circ + \mathbf{k} \cos 40^\circ \sin 45^\circ)$$

$$\mathbf{M}_{500} = 18.95\mathbf{i} - 5.59\mathbf{j} - 16.90\mathbf{k} \text{ N}\cdot\text{m}$$

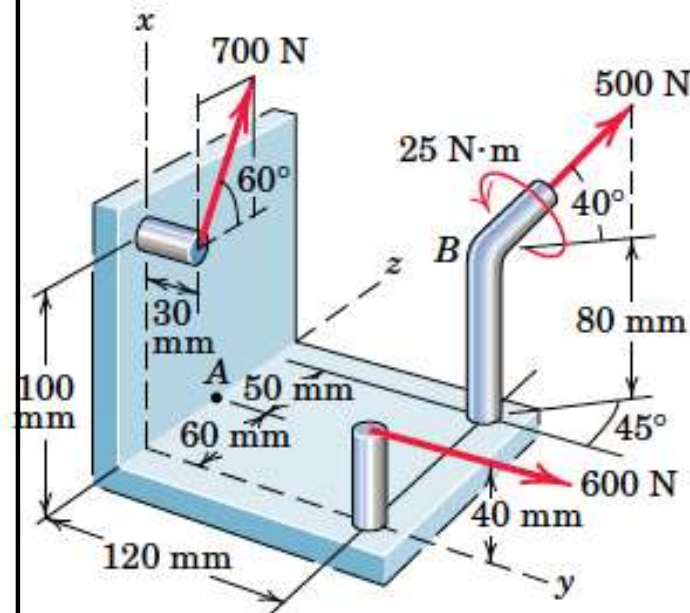
$$\begin{aligned} \mathbf{M}_{600} &= (600)(0.060)\mathbf{i} + (600)(0.040)\mathbf{k} \\ &= 36.0\mathbf{i} + 24.0\mathbf{k} \text{ N}\cdot\text{m} \end{aligned}$$

$$\begin{aligned} \mathbf{M}_{700} &= (700 \cos 60^\circ)(0.030)\mathbf{i} - [(700 \sin 60^\circ)(0.060) \\ &\quad + (700 \cos 60^\circ)(0.100)]\mathbf{j} - (700 \sin 60^\circ)(0.030)\mathbf{k} \\ &= 10.5\mathbf{i} - 71.4\mathbf{j} - 18.19\mathbf{k} \text{ N}\cdot\text{m} \end{aligned}$$

$$\begin{aligned} \mathbf{M}' &= 25.0(-\mathbf{i} \sin 40^\circ - \mathbf{j} \cos 40^\circ \cos 45^\circ - \mathbf{k} \cos 40^\circ \sin 45^\circ) \\ &= -16.07\mathbf{i} - 13.54\mathbf{j} - 13.54\mathbf{k} \text{ N}\cdot\text{m} \end{aligned}$$

$$\mathbf{M} = 49.4\mathbf{i} - 90.5\mathbf{j} - 24.6\mathbf{k} \text{ N}\cdot\text{m}$$

$$M = \sqrt{(49.4)^2 + (90.5)^2 + (24.6)^2} = 106.0 \text{ N}\cdot\text{m}$$



THANK YOU